

Vaibhav Jayakumar

Mechanical Engineering @ Cal Poly SLO

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Education

- California Polytechnic University at San Luis Obispo | B.S. in Mechanical Engineering | Graduation: 2029
- Irvington High School | Unweighted GPA: 3.98 | Graduation: 2025
- Relevant Coursework: Linear Analysis, Intro. to Engineering Design, Principles of Engineering, Subtractive Manufacturing, Physics (Mechanics, Waves & Optics, Electricity & Magnetism), AP Computer Science A, Materials Joining, Intro to ME

Skills

- Technical: Autodesk Fusion 360, Autodesk Inventor, Onshape, Adobe Illustrator, PRUSA Slicer, Python, Java
- Soft: Public speaking, Leadership, Problem Solving, Communication

Leadership & Work Experience

Mechanical Engineering Intern | Food Tech Startup in Stealth Mode | Fremont, CA June 2024 - July 2025

- Manufactured components for 20+ egg cooking robots utilizing CNCs, SLA printers, drill presses, and table saws
- Rapid prototyping and testing: designed electrical components and soldered PCBs to optimize functionality
- Constructed a dynamometer to test motor torque, ensuring efficient performance
- Conducted performance tests on electronic components, iterating designs for improved reliability
- Optimized mechanism efficiency by refining blade angles, arm movements, and hot plate designs

Team Member, Mechanical Lead, Team Captain | First Tech Challenge (FTC) | Fremont, CA Aug 2020 - Jan 2024

- Team Captain - Techarinos 18767 (2020 - 2021)
 - Led the mechanical design team to create and design a 3D-printed claw to grab a "Wobble Goal"
 - Led an outreach initiative to teach over 120 kids Scratch and basic programming
- Created, produced, and constructed effectively packaged mechanisms
- Ranked top 18th FTC team by FUN Robotics Network (Future 14473) (2022)
- Team Captain, Mechanical Lead - Blizzard Robotics 18373 (2023)
 - Led a team of 8 to develop a robot for the 2023 FTC Centerstage season

Entrepreneur in Residency & School Representative | Pilotcity Inc. | Fremont, CA Mar 2023 - July 2025

- Developed a 3D-printed engineering kit: a reusable, disassemblable, lightweight flywheel-powered toy car
- Manufactured 25+ kits and distributed them internationally
- Secured 23 school/teacher partnerships for PilotCity enrollment
- Mentored 9 interns and organized weekly check-ins, ensuring successful program completion

Lead STEM Instructor & Student Engagement Leader | TinkerOn Labs | Fremont, CA May 2024 - July 2025

- Co-developed curriculum and taught STEM concepts to 60+ elementary students through hands-on classes
- Supervised students during all activities and enforced safety practices
- Led hands-on STEM demos for 200+ kids at Fremont Science Night

Intern & Volunteer (Coach, Fundraiser) | African Library Project June 2017 - July 2025

- Led 17 book drives across 7 countries; donated 15,000+ books and raised \$7,500

Honors & Awards

- 2024 Gold Presidential Volunteer Service Award - 250+ hours for the African Library Project
- 2023 Irvington High School PLTW Principles of Engineering Award of Excellence
- [2023 Runner-up, Instructables Contest: Featured on Homepage for robotic air hockey table](#)
- 2023 FTC Awards - NorCal Regionals Motivate, South Bay League Motivate, and Third Inspire
- 2021 FTC Awards - Design and Innovate